

Grand Council — Session of 2026-07-18

Convened by James Gianotti · Space: Grand Council · Synthesist: Perplexity Computer ·
Backing models: Physicist gpt_5_5 · Engineer claude_sonnet_4_6 · Mathematician
gemini_3_1_pro · Historian-Philosopher claude_opus_4_8 · Skeptic gemini_3_1_pro

Docket

Artifact A. Lorenz Equation infographic · tier: Textbook · **Ruling: Sound**

Artifact B. Sam Mathverse matrix series (3 pages) · tier: Textbook · Ruling: Reference

Artifact C. PKM-BLT “Cosmic Evolution Model” · tier: Self-published · **Ruling: Reject**

Artifact D. 1000W ZVS induction heater render · tier: Hobbyist · **Ruling: Mixed**

Artifact E. Volumetric Flow / Continuity Equation · tier: Textbook · **Ruling: Sound***

Artifact A — Lorenz Equation infographic

Intake. Pedagogical card summarizing Lorenz’s 1963 three-ODE system, canonical parameters ($\sigma=10$, $\rho=28$, $\beta=8/3$), the butterfly attractor, and its role in chaos theory. Evidence tier: Textbook.
Claims Under Review: C1–C6.

Physicist. ODEs and parameter roles (C1, C2) are stated correctly; deterministic sensitive dependence and two-lobed strange attractor (C3, C4) are standard. Only C5 needs qualification — the “butterfly” title of the 1972 AAAS talk is often credited to Philip Merilees, not Lorenz.

Engineer. I abstain — outside my charge.

Mathematician. Dissipative system: $\text{div } F = -(\sigma+1+\beta) < 0$, so phase-space volumes contract. “Fractal / figure-eight” (C4) is technically imprecise; the attractor is a *strange attractor* with Hausdorff dimension ≈ 2.06 .

Historian-Philosopher. “Father of Chaos Theory” (C6) is defensible but should be historicized — real precursors include Poincaré (1890 three-body) and Smale’s horseshoe (early 1960s). Lorenz is a founder, not the sole father. C5 correction stands — Merilees named the 1972 talk (earlier metaphor was a seagull).

Skeptic’s cross. C5 is historically simplistic but popularly accepted. No Motion to Reject. Sound reference material.

Ruling — Sound. The card transmits the canonical Lorenz system and its consequences faithfully. Only C5 (naming attribution) requires an asterisk. Everything else is right.

Action. recursive-state-dynamics :: implement a Lorenz($\sigma=10$, $\rho=28$, $\beta=8/3$) reference trajectory in the L/K/E simulator and compare its Lyapunov spectrum to RSD test cases :: reproduce $\lambda_1 \approx 0.906$, $\lambda_2 \approx 0$, $\lambda_3 \approx -14.57$ to within 5%

Artifact B — Sam Mathverse matrix series (three pages)

Intake. Sequential handwritten-style linear-algebra reference pages (sections 3–15) covering equal matrices, transpose, operations, zero/identity, diagonal, scalar, triangular, symmetric, skew-symmetric, Hermitian, and skew-Hermitian matrices. Evidence tier: Textbook. Claims Under Review: C1–C3.

Physicist. I abstain — outside my charge.

Engineer. I abstain — outside my charge.

Mathematician. Audited every definition. $(A^T)^T = A$ ✓; symmetric $a_{ij} = a_{ji}$ ✓; skew $a_{ii} = 0$ ✓; Hermitian $A^* = A$ with real diagonal ✓; skew-Hermitian with real entries reduces to skew-symmetric ✓; multiplication $c_{ij} = \sum_k a_{ik} b_{kj}$ with $(m \times n)(n \times p) = m \times p$ ✓. All definitions and derived properties are rigorously correct.

Historian-Philosopher. Lineage runs through Cayley's 1858 *Memoir on the Theory of Matrices* and Hermite's mid-19th-c. forms with real spectra. Faithful transmission of a 170-year-old canon. Only meaning-question: no attribution to Cayley or Hermite on the cards themselves.

Skeptic's cross. No claims overstated. No Motion to Reject.

Ruling — Reference. Correct standard linear-algebra pedagogy. No dispute to litigate. Attribution to Cayley/Hermite would strengthen the deck.

Action. None — Reference material. Optional: add a fourth card citing Cayley (1858) and Hermite (~1855) for provenance.

Artifact C — PKM-BLT “Cosmic Evolution Model”

Intake. A yellow single-page infographic claiming to be a “COMPLETE MATHEMATICAL FRAMEWORK OF THE UNIVERSE” built on H_0 as a “locked initial value” and the identity $H_0 \cdot t_u = 1$.

Evidence tier: Self-published. Claims Under Review: C1–C7.

Physicist. Reject the physics. $H_0 \cdot t_u = 1$ is not a fundamental identity — it is a tautology because t_u is defined as $1/H_0$ (C1). $A_0 = cH_0$ has units of acceleration but numerical coincidence with MOND’s $a_0 \approx cH_0$ (Milgrom 1983) requires attribution and mechanism, not numerology (C2). $H(t) = H_0(1 - e^{-t/t_u})$ is cosmologically wrong: Λ CDM has $H(t) \rightarrow \infty$ toward the Big Bang and $\rightarrow$ constant only in the late de-Sitter limit (C3, C5). Eleven significant figures on H_0 (C7) is indefensible given the $\sim 2\%$ Planck/SH0ES tension (67.4 vs 73.0 km/s/Mpc).

Engineer. Flag on C7 as a false-precision signal. Nobody who understands uncertainty propagation quotes a $\sim 2\%$ -uncertain quantity to 11 figures.

Mathematician. $\lambda = A_0/c = H_0 = 1/t_u$ is definitions and symbol introductions, not derivations. $H_0 \cdot t_u = 1$ is an algebraic tautology, not an identity. $R(t) = c[t - t_u(1 - e^{-t/t_u})]$ is dimensionally correct but presented without physical interpretation. A framework requires grounded axioms, not cyclic definitions.

Historian-Philosopher. Belongs to the social-media Theory-of-Everything genre — yellow background, master equation, locked constant, tautological identity, invocation of standard textbook relations as if original. Compared with Wolfram’s Physics Project, Weinstein’s Geometric Unity, or Hamein’s work, PKM-BLT sits at the low-rigor end. The “leak” mechanism is undefined; “cosmic controller” is not physics vocabulary. Repackaged mysticism, not lineage.

Skeptic’s cross. Motion to Reject filed on C5. Unfalsifiable “leak” mechanism, no observational predictions distinguishable from Λ CDM, tautological identity, false precision to 11 figures during an active Hubble tension.

Ruling — Reject. Motion to Reject accepted. The four seats agree: PKM-BLT dresses a definition ($t_u := 1/H_0$) as a discovery, contradicts observed cosmic-expansion history, and quotes precision no measurement supports. It does not supersede or subsume Λ CDM. Do not carry it forward.

Action. gravitational-compass :: write a one-page rebuttal note titled “ $H_0 \cdot t_u = 1$ is a definition, not an identity” for docs/, contrasting PKM-BLT $H(t) = H_0(1 - e^{-t/t_u})$ against Λ CDM Friedmann $H(a)$ with Planck 2018 parameters :: rebuttal reproduces Planck Λ CDM $H(z)$ to visual accuracy on a log- z plot from $z=0$ to $z=1100$

Artifact D — 1000W ZVS-style induction heater schematic

Intake. A stylized render of a Mazzilli-family ZVS driver labeled 1000W: 6-turn 6mm work coil, $\sim 2\mu\text{F}$ tank, dual IRFP250N cross-coupled MOSFETs, $100\mu\text{H}/10\text{A}$ chokes, asymmetric 12V/22V Zener gate clamp, 12–48V DC input, no cooling or snubbers shown. Evidence tier: Hobbyist. Claims Under Review: C1–C6.

Physicist. Topology is a valid resonant tank; $\omega_o = 1/\sqrt{LC}$ with $L \approx 1\text{--}2\ \mu\text{H}$ and $C \approx 2\ \mu\text{F}$ gives $f_o \approx 80\text{--}160\ \text{kHz}$ (C1). At 48V, 1kW $\rightarrow$ 21A rail current $\rightarrow I^2R \approx 37\text{W}$ per FET before switching, diode, choke, and tank losses. Heatsink-and-airflow territory (C2, C5). Asymmetric 12/22V gate clamps, 10A chokes, and absent cooling make C3, C4, C6 unsupported for continuous 1kW.

Engineer. Build case fails on C2–C6. IRFP250N $\theta_{jc} \approx 0.64^\circ\text{C/W}$ $\rightarrow$ without heatsinking, T_j hits 175°C in seconds. 22V Zener against IRFP250N absolute-max V_{gs} of $\pm 20\text{V}$ is a direct spec violation. 100 μH /10A chokes saturate long before 20+A rail current; saturated chokes short the rail through the FETs — instant failure. No drain-source snubber; no water cooling on the work coil. Educational render, not a build-from BOM.

Mathematician. Second-order LC oscillator; $Q = (1/R)\sqrt{L/C}$. Canonical Mazzilli topology has two-fold reflection symmetry; the 12V/22V Zener asymmetry (C3) breaks that symmetry — either the render is wrong or the design is wrong.

Historian-Philosopher. Lineage is authentic — Vladimiro Mazzilli's ZVS design, mid-2000s hobbyist forums, originally for flyback / Tesla-coil driving before migrating to induction heating (C1). The render matches Chinese-kit marketing-image style; the asymmetric Zeners are almost certainly an illustration error rather than deliberate design.

Skeptic's cross. Motion to Reject filed on C2 — "1000 W" as stated with this BOM and no cooling/snubbers is dangerously false. Falsifiability test: build it, run 48V continuous; predict thermal shutdown or MOSFET failure within 60 s.

Ruling — Mixed. Topology (C1) is Sound; the claimed 1000W operating point (C2, C5, C6) is Rejected at this BOM; the asymmetric Zener design (C3) is either a render error or a spec violation — treat as render error and build symmetric 12V or 15V. Motion to Reject accepted, narrowed to C2.

Action. cbjg-benchtop-motor-build :: adapt the Mazzilli ZVS topology into a bench prototype at a safer 100–200W design point using IRFP250N with proper heatsinking, symmetric 15V Zeners, saturated-current-rated chokes ($\geq I_{pk} \times 2$), and an RC drain-source snubber :: prototype delivers rated power for ≥ 5 min continuous with FET case $< 80^\circ\text{C}$ on an ambient-air heatsink

Artifact E — Volumetric Flow Rate / Continuity Equation

Intake. Hand-drawn engineering notebook page by Sanjiv Ganjir teaching $Q = A \cdot v$ and $A_1 v_1 = A_2 v_2$ via a garden-hose analogy and a highway-traffic analogy. Evidence tier: Textbook. Claims Under Review: C1–C4.

Physicist. $Q = A \cdot v$ (C1) and $A_1 v_1 = A_2 v_2$ (C2) are correct in the incompressible, one-dimensional, uniform-velocity limit. Mass conservation (C3) is exactly right — the mass-continuity PDE collapses to $\text{div } v = 0$ for constant ρ . Garden-hose intuition honest. Highway analogy (C4) breaks: real traffic is compressible in the LWR sense, so a narrower road lowers velocity and raises density.

Engineer. Sound pedagogy on C1–C3; applications panel is well-chosen. Highway analogy (C4) works for the intuition “throughput conserved,” but any student who uses it for real pipe design will over-predict velocity in narrow sections and under-predict pressure drop. Use the water figure for calculations; discard the highway figure after first exposure.

Mathematician. Formulation is the $\text{div } v = 0$ limit of the mass-continuity PDE. Real traffic per LWR has density-dependent velocity; density spikes at constrictions, velocity drops, backward-propagating shockwaves form — highway flow explicitly violates $\text{div } v = 0$. Equating incompressible flow to macroscopic traffic is structurally false (C4).

Historian-Philosopher. C1–C3 sit atop a ~270-year lineage — Euler (1757), Cauchy’s continuum formalization. Hydraulic-analogy pedagogy (Kirchhoff via water pipes) is a 19th-century tradition. C4 has its own real lineage in Lighthill–Whitham (1955) and Richards (1956), but that lineage makes the analogy technically inexact: LWR traffic is compressible with shockwaves incompressible continuity cannot produce.

Skeptic’s cross. C4 is misleading past introductory framing. No Motion to Reject — the physics stands. Falsifiability test: observe merging lanes; velocity falls as density rises, opposite to incompressible flow.

Ruling — Sound. Core physics (C1–C3) transmitted correctly; pedagogy honest. Highway analogy (C4) should be annotated pedagogically useful, technically inexact — real traffic obeys LWR, not incompressible continuity.

Action. force-carriers :: when preparing the “conservation laws” companion card, use the garden-hose figure as exemplar and explicitly contrast the highway analogy against LWR to inoculate students against the analogy failure mode :: card includes a side-by-side ($\text{div } v = 0$) vs ($\partial \rho / \partial t + \partial(\rho v) / \partial x = 0$) comparison in ≤ 150 words

Cross-cutting synthesis

Faithful pedagogy (A, B, E). Correctly transmit real canonical material. A carries a naming asterisk; E carries an analogy caveat; B is unblemished reference. Council posture: transmit forward.

Legitimate topology, dishonest labeling (D). The Mazzilli ZVS lineage is real and buildable at modest power. The specific render dresses it as a 1000W finished product it is not. Council posture: strip the label, keep the topology, build symmetric and small before scaling.

Definition-dressed-as-discovery (C). PKM-BLT is a tautology ($t_u := 1/H_o$) wrapped in yellow with numerology-tier precision, contradicting the observed $H(t)$ profile. Council posture: reject and rebut. Do not let this pattern contaminate the gravitational-compass work.

Portfolio impact

- recursive-state-dynamics — gain a Lorenz Lyapunov-spectrum validation harness (from A).
- gravitational-compass — gain a written rebuttal against the PKM-BLT genre (from C).
- cbjg-benchtop-motor-build — gain a safer, symmetric Mazzilli ZVS prototype target (from D).
- force-carriers — gain an LWR-vs-continuity contrast card (from E).

The user is the emperor of the earth. Any ruling may be overturned with a stated reason; overrides are logged as addenda. Source repository: github.com/giomj/dev/skills/grand-council.